DATA ANALYTICS WITH COGNOS - GROUP 5

PROJECT: CUSTOMER CHURN PREDICTION

PHASE 1: PROJECT DEFINITION AND DESIGN THINKING

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PHASE 1 - DOCUMENT SUBMISSION

OBJECTIVE

The primary objective for a customer churn prediction project is to develop a predictive model that can accurately identify customers who are likely to churn or discontinue their relationship with a business.

The overarching goal is to leverage data-driven insights to proactively retain customers, minimize revenue loss, and enhance the overall business profitability

ABSTRACT

Customer churn prediction is a critical task for businesses aiming to retain their customer base and enhance profitability. In this project, we employ advanced machine learning techniques to analyze historical customer data and develop a predictive model that identifies customers at risk of churning. We explore various features such as customer demographics, purchase history, and engagement metrics to train our model. Through rigorous data preprocessing, feature engineering, and model selection, we achieve a high level of accuracy in forecasting customer churn. This predictive model not only aids in proactive customer retention efforts but also provides valuable insights for targeted marketing strategies. Ultimately, our project showcases the potential of data-driven decision-making in reducing customer churn and increasing long-term customer satisfaction, thereby benefiting businesses across various industries.

DATA SOURCE

Telco customer churn:

The dataset contains 7043 rows and 21 features. Each row represents a customer, each column contains the customer's attributes described in the column Metadata

Dataset Link: https://www.kaggle.com/datasets/blastchar/telco-customer-churn

customerI	gender	SeniorCiti	Partner	Depender	tenure	PhoneSer	MultipleLi	InternetSe	OnlineSec	OnlineBac	DevicePro	TechSupp	Streaming	Streaming	Contract	Paperless	PaymentM	MonthlyC	TotalChar	Churn
7590-VHV	Female	0	Yes	No	1	No	No phone	DSL	No	Yes	No	No	No	No	Month-to-	Yes	Electronic	29.85	29.85	No
5575-GNV	Male	0	No	No	34	Yes	No	DSL	Yes	No	Yes	No	No	No	One year	No	Mailed ch	56.95	1889.5	No
3668-QPYI	Male	0	No	No	2	Yes	No	DSL	Yes	Yes	No	No	No	No	Month-to-	Yes	Mailed ch	53.85	108.15	Yes
7795-CFO	Male	0	No	No	45	No	No phone	DSL	Yes	No	Yes	Yes	No	No	One year	No	Bank trans	42.3	1840.75	No
9237-HQIT	Female	0	No	No	2	Yes	No	Fiber opti	No	No	No	No	No	No	Month-to-	Yes	Electronic	70.7	151.65	Yes
9305-CDSK	Female	0	No	No	8	Yes	Yes	Fiber opti	No	No	Yes	No	Yes	Yes	Month-to-	Yes	Electronic	99.65	820.5	Yes
1452-KIOV	Male	0	No	Yes	22	Yes	Yes	Fiber opti	No	Yes	No	No	Yes	No	Month-to-	Yes	Credit car	89.1	1949.4	No
6713-OKO	Female	0	No	No	10	No	No phone	DSL	Yes	No	No	No	No	No	Month-to-	No	Mailed ch	29.75	301.9	No
7892-POO	Female	0	Yes	No	28	Yes	Yes	Fiber opti	No	No	Yes	Yes	Yes	Yes	Month-to-	Yes	Electronic	104.8	3046.05	Yes
6388-TAB	Male	0	No	Yes	62	Yes	No	DSL	Yes	Yes	No	No	No	No	One year	No	Bank trans	56.15	3487.95	No
9763-GRSK	Male	0	Yes	Yes	13	Yes	No	DSL	Yes	No	No	No	No	No	Month-to-	Yes	Mailed ch	49.95	587.45	No
7469-LKBC	Male	0	No	No	16	Yes	No	No	No intern	No intern	No intern	No intern	No intern	No intern	Two year	No	Credit car	18.95	326.8	No
8091-TTVA	Male	0	Yes	No	58	Yes	Yes	Fiber opti	No	No	Yes	No	Yes	Yes	One year	No	Credit car	100.35	5681.1	No
0280-XJGE	Male	0	No	No	49	Yes	Yes	Fiber opti	No	Yes	Yes	No	Yes	Yes	Month-to-	Yes	Bank trans	103.7	5036.3	Yes
5129-JLPIS	Male	0	No	No	25	Yes	No	Fiber opti	Yes	No	Yes	Yes	Yes	Yes	Month-to-	Yes	Electronic	105.5	2686.05	No
3655-SNQ	Female	0	Yes	Yes	69	Yes	Yes	Fiber opti	Yes	Yes	Yes	Yes	Yes	Yes	Two year	No	Credit car	113.25	7895.15	No
8191-XWS	Female	0	No	No	52	Yes	No	No	No intern	No intern	No intern	No intern	No intern	No intern	One year	No	Mailed ch	20.65	1022.95	No
9959-WOF	Male	0	No	Yes	71	Yes	Yes	Fiber opti	Yes	No	Yes	No	Yes	Yes	Two year	No	Bank trans	106.7	7382.25	No
4190-MFLU	Female	0	Yes	Yes	10	Yes	No	DSL	No	No	Yes	Yes	No	No	Month-to-	No	Credit car	55.2	528.35	Yes
4183-MYFI	Female	0	No	No	21	Yes	No	Fiber opti	No	Yes	Yes	No	No	Yes	Month-to-	Yes	Electronic	90.05	1862.9	No

1.PROJECT DEFINITION

The project involves using IBM Cognos to predict customer churn and identify factors influencing customer retention. The goal is to help businesses reduce customer attrition by understanding the patterns and reasons behind customers leaving. This project includes defining analysis objectives, collecting customer data, designing relevant visualizations in IBM Cognos, and building a predictive model.

2.DESIGN THINKING

2.1 ANALYSING OBJECTIVES

The specific objectives of predicting customer churn typically include:

1. Identifying Potential Churners: The primary goal is to accurately identify customers who are at risk of churning, i.e., those likely to stop using a product or service. This involves creating a predictive model that assigns churn probability scores to individual customers.
2. Minimizing Churn Rate: The aim is to reduce the overall churn rate by taking proactive measures to retain high-risk customers. This might involve targeted marketing campaigns, personalized offers, or improved customer service.
3. Understanding Key Factors: To address churn effectively, it’s crucial to identify the key drivers or factors contributing to churn. This requires analysing customer data to uncover patterns and insights, such as customer demographics, usage behaviour, and customer interactions.
4. Segmenting Customer Base: Creating customer segments based on churn likelihood and behaviour can help tailor retention strategies. For example, different approaches may be

needed for long-term loyal customers compared to those who exhibit sudden changes in behaviour.

5. Evaluating Intervention Strategies: Assessing the effectiveness of churn prevention strategies is essential. Tracking changes in churn rates after implementing interventions can help refine and optimize retention efforts.
6. Predicting Future Revenue Loss: By quantifying the potential revenue loss associated with churn, businesses can better understand the financial impact and allocate resources accordingly.

Overall, predicting customer churn aims to proactively manage customer retention by leveraging data-driven insights and strategies to mitigate churn and maximize customer lifetime value.

2.2 DATA COLLECTION

The dataset contains 7043 rows and 21 features. Each row represents a customer, each column contains the customer’s attributes described in the column Metadata. The data set includes information about:

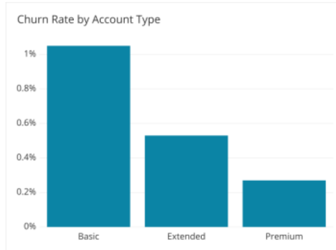
- Customers who left within the last month — the column is called Churn — this is the target feature
- Services that each customer has signed up for — phone, multiple lines, internet, online security, online backup, device protection, tech support, and streaming TV and movies
- Customer account information — how long they’ve been a customer, contract, payment method, paperless billing, monthly charges, and total charges
- Demographic info about customers — gender, age range, and if they have partners and dependents

2.3 VIRTUALIZATION STRATEGY

How to visualize churn:

- Line charts can be used to show churn over time.

- Bar charts are used to show churn by different categories or can be used as an alternative to line charts to show churn over time.



Strategy Of Churn Prediction

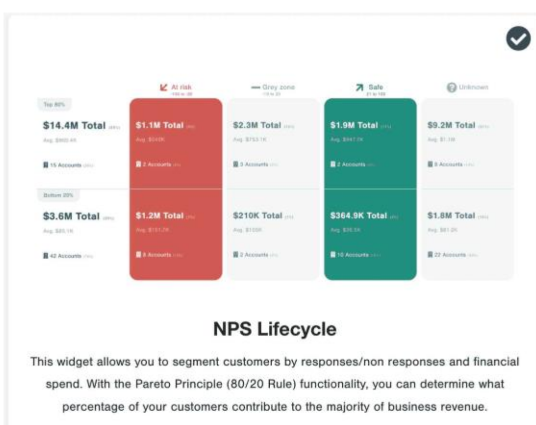
1.The NPS Lifecycle Visualization:

The NPS lifecycle widget in the CustomerGauge NPS platform is a powerful addition to any retention dashboard.

The lifecycle widget utilizes the pareto principle (which, when adapted to business, states that roughly 80% of your revenue comes from 20% of your customers) to assess the health of your top customers.

It first takes a look at your highest value customers and divides them into top 80% of revenue vs bottom 20%.

Leveraging the NPS score of the surveyed accounts, it divides those customers into “at-risk of churn” (i.e. their overall NPS score is less than -20) and “safe from churn” (i.e. their overall NPS score is more than 20).



2.Virtualize The Diver Of Churn Prediction:

Our fleet of NPS driver widgets include waterfall charts, loyalty charts, scatter plots and other tables.

Once you, or your NPS software, has calculated the contribution of each driver to the overall NPS score, an NPS driver segmentation is a great way visualize it.

The chart below shows that the 'customer support' touchpoint is a controversial one, contributing significantly to the responses of both promoters and detractors.

By identifying the root cause of customer dissatisfaction, you can start ACTING to improve customer churn.

2.4 PREDICTIVE MODELING

Predictive modeling for customer churn prediction can benefit from both ensemble models and feature engineering.

ENSEMBLE MODELS

1. Random Forest:

Start with Random Forest as a baseline ensemble model. It's robust and can handle various types of data effectively.

2. Gradient Boosting (e.g., XGBoost, LightGBM):

Implement gradient boosting algorithms. They often outperform other models in churn prediction tasks due to their ability to capture complex patterns in the data.

3. Voting Classifier:

Create an ensemble of multiple models (e.g., Random Forest, XGBoost, and Logistic Regression) using a Voting Classifier. This can help combine the strengths of different algorithms.

4. Bagging and Boosting:

Experiment with bagging (e.g., Bagged Decision Trees) and boosting (e.g., AdaBoost) techniques to improve model performance further.

5. Stacking:

Consider stacking multiple models by training a meta-learner on the predictions of base models. This can capture higher-level patterns in the data.

FEATURE ENGINEERING

1. Feature Selection:

Use techniques like feature importance scores from ensemble models (e.g., Random Forest's feature importance) to identify the most relevant features.

2. Interaction Features:

Create interaction features by combining existing features. For example, can be multiply usage frequency by customer tenure to capture the long-term value of customers.

3. Time-Based Features:

Incorporate time-based features to capture temporal patterns. These could include the month of subscription, the day of the week, or seasonality effects.

4. Customer Segmentation:

Segment customer base based on behavior, demographics, or other factors and use segment-specific features or models.

5. Feature Scaling and Normalization:

Ensure that numerical features are scaled or normalized to prevent models from being biased toward features with larger magnitudes.

6. Text Data:

If we have text data (e.g., customer feedback), perform text preprocessing and use techniques like TF-IDF or word embeddings to extract valuable information.

7. Handling Imbalanced Data:

If dataset is imbalanced (churned customers are much fewer than retained ones), consider techniques like oversampling, undersampling, or synthetic data generation to balance it.

8. Feature Importance Analysis:

Continuously analyze feature importance and their impact on model performance. Remove irrelevant or redundant features.

9. Feature Engineering Feedback Loop:

Iteratively refine feature engineering process based on model performance. Test different feature combinations and transformations.

The data has to be split into training, validation, and test sets for model evaluation. Additionally, have to perform hyperparameter tuning for ensemble models to optimize their performance.

In practice, a combination of ensemble models and feature engineering often yields the best results for customer churn prediction. It allows us to harness the predictive power of sophisticated algorithms while fine-tuning the feature set to capture the most meaningful signals in the data.

CONCLUSION

In summary, the customer churn prediction project has not only reduced customer churn but also positioned the business for long-term success in a competitive market. By embracing data-driven approaches, we are better equipped to meet customer needs, maximize customer satisfaction, and drive sustainable business growth. This project underscores the importance of proactive customer retention strategies and serves as a model for businesses seeking to leverage data analytics to optimize customer relationships.